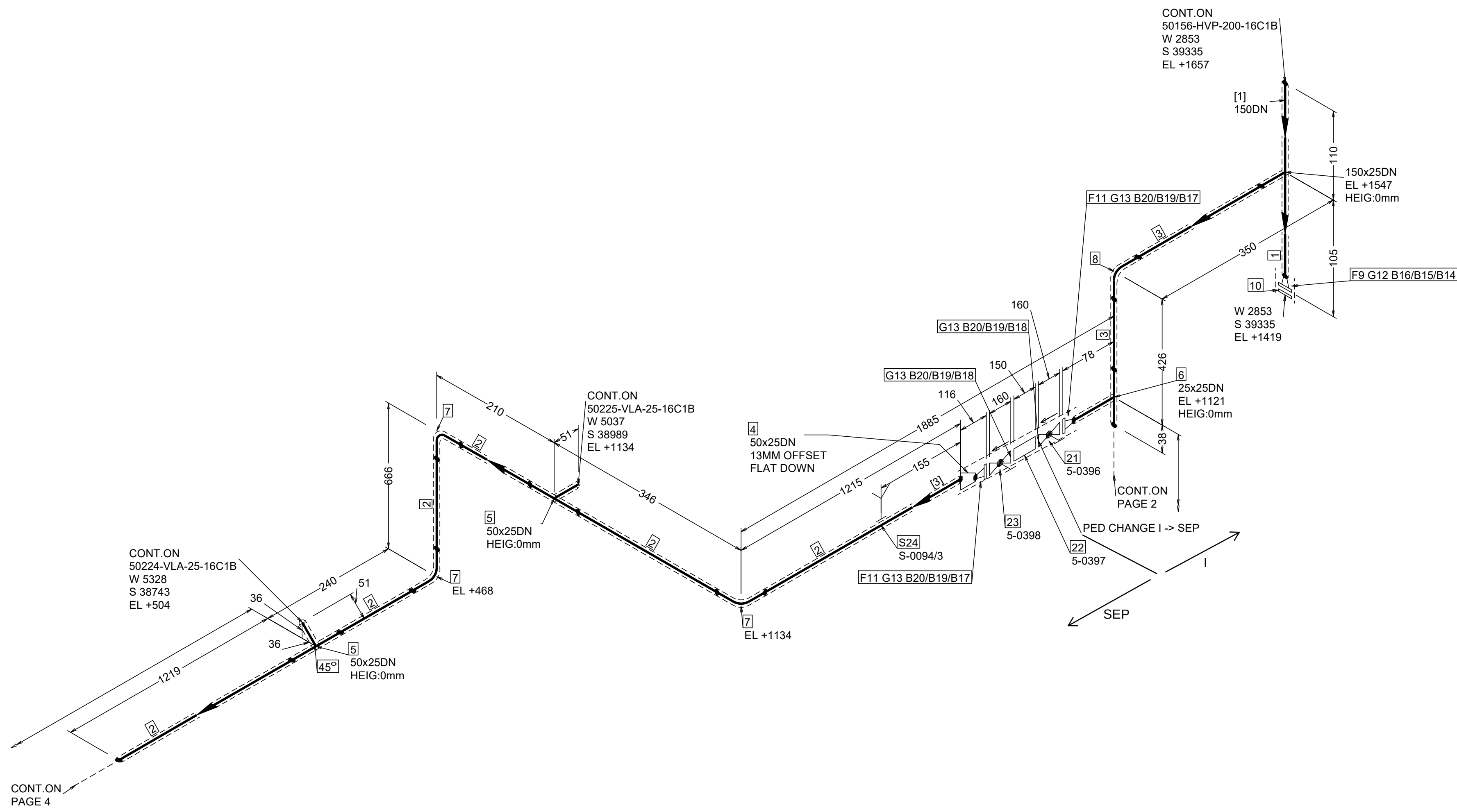


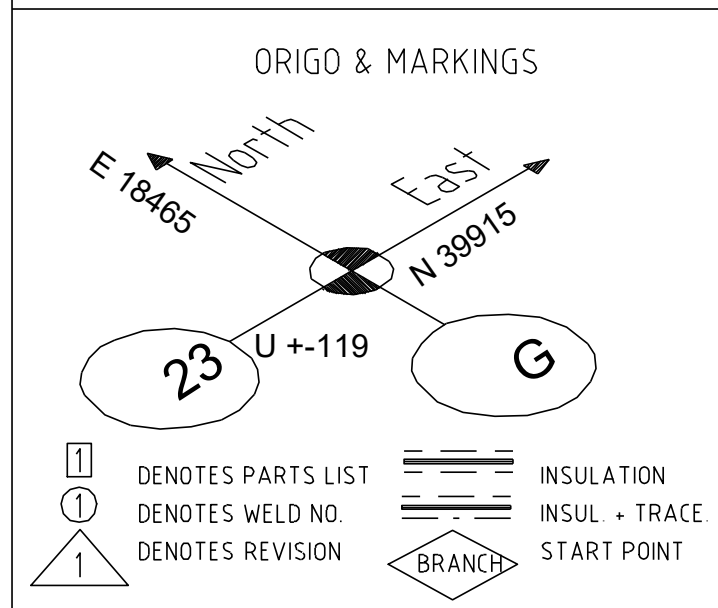
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NO	PART DESCRIPTION	DN (MM)	PIPE SPEC	QUANTITY
1	Tube - 168.3 x 4.50 - EN 10216-2 - P235GH-TC1	150	119440-VE16C1C-R00	160MM
2	Tube - 60.3 x 2.90 - EN 10216-2 - P235GH-TC1	50	119440-VE16C1C-R00	3184MM
3	Tube - 33.7 x 2.60 - EN 10216-2 - P235GH-TC1	25	119440-VE16C1C-R00	580MM
4	Eccentric reducer - EN 10253-2 - type A - 60.3 x 2.90 / 33.7 x 2.60 - P235GH-TC1	50 x 25	119440-VE16C1C-R00	1
5	Reduced tee - EN 10253-2 - type A - 60.3 x 2.90 / 33.7 x 2.60 - P235GH-TC1	50 x 25	119440-VE16C1C-R00	2
6	Equal tee - EN 10253-2 - type A - 33.7 x 2.60 - P235GH-TC1	25 x 25	119440-VE16C1C-R00	1
7	Elbow - EN 10253-2 - type A - model 3D - 90deg - 60.3 x 2.90 - P235GH-TC1	50	119440-VE16C1C-R00	3
8	Elbow - EN 10253-2 - type A - model 3D - 90deg - 33.7 x 2.60 - P235GH-TC1	25	119440-VE16C1C-R00	1
9	Weld-neck flange - EN 1092-1 - type 11 - B1 - 168.3/DN150 - PN16 - P250GH	150	119440-VE16C1C-R00	1
10	Blind flange - EN 1092-1 - type 05 - B1 - 168.3/DN150 - PN16 P265GH	150	119440-VE16C1C-R00	1
11	Weld-neck flange - EN 1092-1 - type 11 - B1 - 33.7/DN25 - PN40 - P250GH	25	119440-VE16C1C-R00	2
NO	PART DESCRIPTION	DN (MM)	PIPE SPEC	QUANTITY
12	Gasket - EN 1514-1 - type IBC - OD168.3/DN150 - 1.5mm Thk - PN16 Fiber	150	119440-VE16C1C-R00	1
13	Gasket - EN 1514-1 - type IBC - OD33.7/DN25 - 1.5mm Thk - PN40 Fiber	25	119440-VE16C1C-R00	4
14	80 mm Bolt EN ISO 4014 25CrMo4, EN 10269, mat cert. 3.1, HDG	20		8
15	Hexagon nut EN ISO 4032 25CrMo4, EN 10269, mat cert. 3.1, HDG	20		8
16	Washer EN ISO 7089 Grade A - S235JR EN10025-2 - HDG	20		16
17	60 mm Bolt EN ISO 4014 25CrMo4, EN 10269, mat cert. 3.1, HDG	12		8
18	55 mm Bolt EN ISO 4014 25CrMo4, EN 10269, mat cert. 3.1, HDG	12		8
19	Hexagon nut EN ISO 4032 25CrMo4, EN 10269, mat cert. 3.1, HDG	12		16
20	Washer EN ISO 7089 Grade A - S235JR EN10025-2 - HDG	12		32
21	BASIC GLOBE VALVE FLANGED PN40 WI HAND WHEEL 5-0396	25	KBE/ALV01	1
22	KPK BASIC FILTER FLANGED PN40 5-0397	25	KBE/ALV01	1
23	BASIC GLOBE VALVE FLANGED PN40 WI HAND WHEEL 5-0398	25	KBE/ALV01	1
	<u>SUPPORTS</u>			
24	SLIDE SUPPORT PSK 7303 TYPE 1, HIGH See supportlist	50	VOFSUPP-PRIM-PSK-AT TA	1

PIPE SPOOLS

[1] [3]



Insulation table (lengths according SSG 7645 AlI2)			
Bore	Thickness	Length	Insulation spec name
25	80mm	2177mm	/PSK3704-4a-good
40	80mm	7431mm	/PSK3704-4a-good
40	40mm	13398mm	/PSK3704-6a-Prot-alum
50	80mm	160mm	/PSK3704-4a-good
50	100mm	7991mm	/PSK3704-4a-good
50	40mm	15293mm	/PSK3704-6a-Prot-alum
150	120mm	407mm	/PSK3704-4a-good

NOTE:

Site weldings to be added by contractor.

Buildings construction and equipment erection tolerances to be checked by contractor.

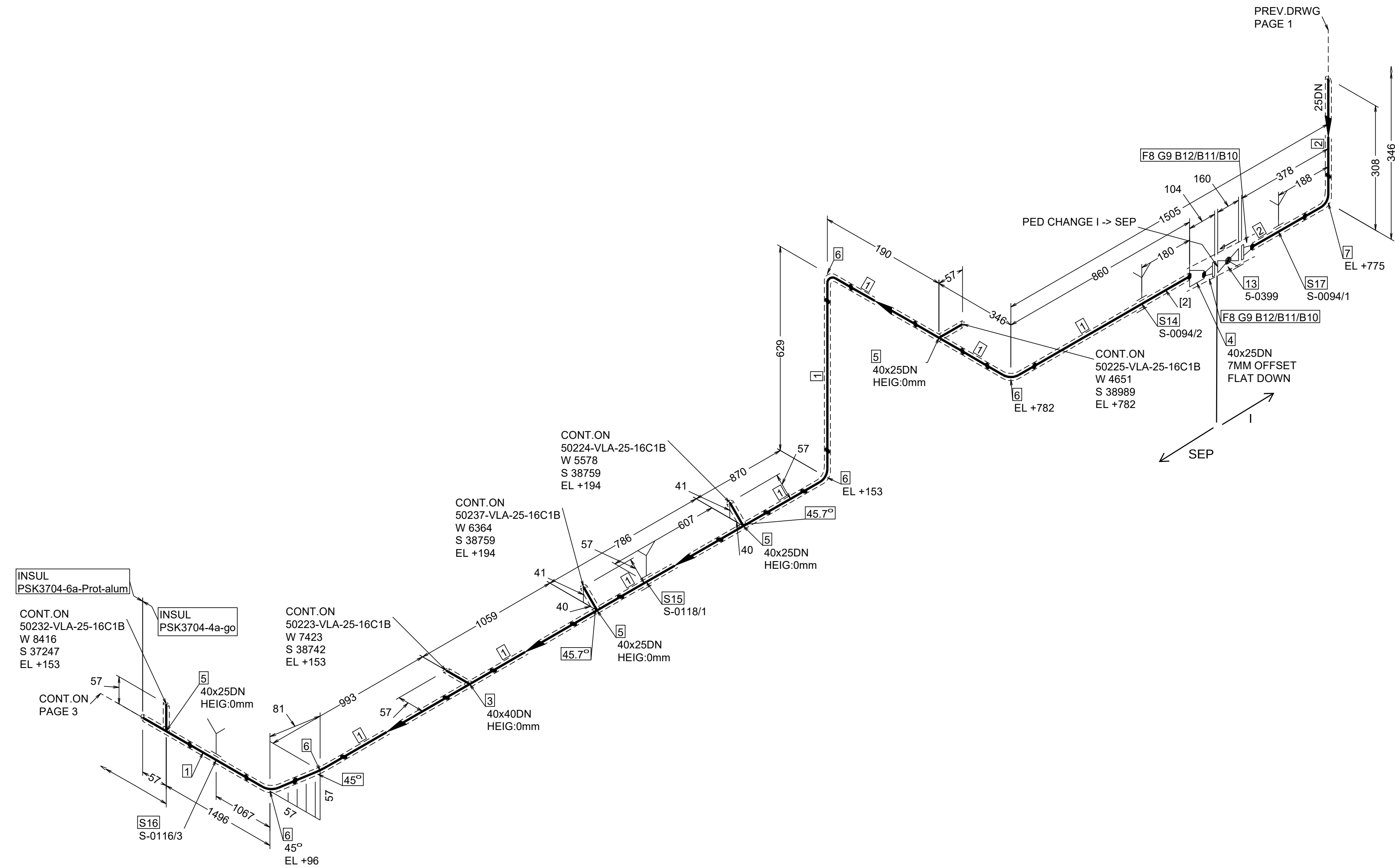
Pipe levels and dimensions are according to pipe center line

Isometric drawings for sizes DN 15-25 are approximate.
Piping contractor is responsible for final routing and supporting

Welding factor	Acc. to pipe class
Efficiency of long. weld	Acc. to pipe class
NDT circ. weld %	-
NDT long. weld %	-
Welding list see doc.	-

		Hazardous CLP content		Legal responsibility to be decided by customer.						
Directive	2014/68/EU	Pipe Class	/119440-VE16C1C-R00	Rev/ Revision Note		Created by		Checked by	Approved by	Date
Design press. (min)	-1bar	PED Category	1/ SEP	0		J. Lakka	T. Saarinen	M. Kankkunen		5/5/2026
Design temp. (min)	-	Media		Site Tervasaari, Valkeakoski		Equipment Type Höyry- ja lauhdejärjestelmä		Project Number NS265A		Equipment ID
Max. allow. press. (PSI)	-	Standard		Project Key TERFMS_NS265A		Ter PM 5 Safe Future				Work
Operating press.	8.5bar	P&ID	RAUAF00303	Projection Method  ISO 5553		General Tolerances ISO 13 920-BF ISO 2768-mK-E ISO 2768-mK-E ISO 2768-mK-E		Scale -	Size A1	Weight -
Operating temp.	198degC	Design press.	10.7bar	 This document is the exclusive intellectual property of Valmet Corporation and/or its subsidiaries (collectively and individually "Valmet") and is furnished solely for executing and maintaining the specific project. Re-use of the document for any other project or purpose is prohibited. The document or the information shall not be reproduced, copied or disclosed to any third party without the prior written consent of Valmet. By accessing this document, you accept these terms of use.						
PED Module	-	Design temp.	250degC	Title 50226-VLA-25-16C1B						
PED Group	-	Test press.	-							
Wind loads	-	Pipe list doc. ID	RAUAF04300	Status CERTIFIED		Document ID				Sheet 1 / 5
External loads	-	Valve list doc. ID	RAUAF04301	Security Classification Internal						Revision 0

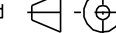
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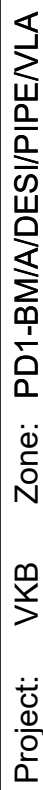


NO	PART DESCRIPTION	DN (MM)	PIPE SPEC	QUANTITY
1	Tube - 48.3 x 2.60 - EN 10216-2 - P235GH-TC1	40	119440-VE16C1C-R00	6293MM
2	Tube - 33.7 x 2.60 - EN 10216-2 - P235GH-TC1	25	119440-VE16C1C-R00	570MM
3	Equal tee - EN 10253-2 - type A - 48.3 x 2.60 - P235GH-TC1	40 x 40	119440-VE16C1C-R00	1
4	Eccentric reducer - EN 10253-2 - type A - 48.3 x 2.60 / 33.7 x 2.60 - P235GH-TC1	40 x 25	119440-VE16C1C-R00	1
5	Reduced tee - EN 10253-2 - type A - 48.3 x 2.60 / 33.7 x 2.60 - P235GH-TC1	40 x 25	119440-VE16C1C-R00	4
6	Elbow - EN 10253-2 - type A - model 3D - 90deg - 48.3 x 2.60 - P235GH-TC1	40	119440-VE16C1C-R00	5
7	Elbow - EN 10253-2 - type A - model 3D - 90deg - 33.7 x 2.60 - P235GH-TC1	25	119440-VE16C1C-R00	1
8	Weld-neck flange - EN 1092-1 - type 11 - B1 - 33.7/DN25 - PN40 - P250GH	25	119440-VE16C1C-R00	2
NO	PART DESCRIPTION	DN (MM)	PIPE SPEC	QUANTITY
9	Gasket - EN 1514-1 - type IBC - OD33.7/DN25 - 1.5mm Thk - PN40 Fiber	25	119440-VE16C1C-R00	2
10	60 mm Bolt EN ISO 4014 25CrMo4, EN 10269, mat cert. 3.1, HDG	12		8
11	Hexagon nut EN ISO 4032 25CrMo4, EN 10269, mat cert. 3.1, HDG	12		8
12	Washer EN ISO 7089 Grade A - S235JR EN10025-2 - HDG	12		16
13	BASIC GLOBE VALVE FLANGED PN40 WI HAND WHEEL 5-0399	25	KBE/VALV01	1
	SUPPORTS			
14	GUIDED SLIDE PSK 7303 TYPE 2, HIGH See supportlist	40	VOFSUPP-PRIM-PSK-AT TA	1
15	GUIDED SLIDE PSK 7303 TYPE 2, LOW See supportlist	40	VOFSUPP-PRIM-PSK-AT TA	1
16	SLIDE SUPPORT PSK 7303 TYPE 1, HIGH See supportlist	40	VOFSUPP-PRIM-PSK-AT TA	1
17	SLIDE SUPPORT PSK 7303 TYPE 1, HIGH See supportlist	25	VOFSUPP-PRIM-PSK-AT TA	1

PIPE SPOOLS

[2]

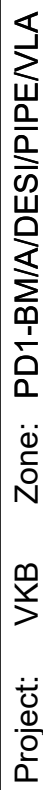
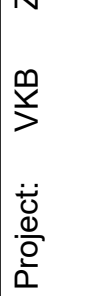
		Hazardous CLP content		Revision 1		Created by		Checked by		Approved by		Date	
Directive	2014/68/EU	Pipe Class	/I19440-VE16C1C-R00	Rev1 Revision Note		J. Lakka		T. Saarinen		M. Kankunen		5/5/2026	
Design press. (min)	-1bar	PED Category	1/ SEP	Site Tervasaari, Valkeakoski		Equipment Type Höyry- ja lauhdejärjestelmä				Equipment ID			
Design temp. (min)	-	Media		Project Key TERPM5_N5265A Ter PM 5 Safe Future		Project Number N5265A				Work			
Max. allow. press. (PS)	-	Standard		Welding Symbols Method  ISO 5553		General Tolerances ISO 13 920-BF ISO 2768-mK-E ISO 2768-mK-E ISO 2768-mK-E		Scale -		Size A1		Weight -	
Operating press.	8.5bar	P&ID	RAUBF00303	 This document is the exclusive intellectual property of Valmet Corporation and/or its subsidiaries (collectively and individually "Valmet") and is furnished solely for executing and maintaining the specific project. Re-use of the document for any other project or purpose is prohibited. The document or the information shall not be reproduced, copied or disclosed to any third party without the prior written consent of Valmet. By accessing this document, you accept these terms of use.									
Operating temp.	198degC	Design press.	10.7bar			Title 502226-VLA-25-16C1B							
PED Module	-	Design temp.	250degC										
PED Group	-	Test press.	-										
Wind loads	-	Pipe list doc. ID	RAUAF04300	Status CERTIFIED		Document ID							
External loads	-	Valve list doc. ID	RAUAF04301	Security Classification Internal		Revision 0							

ABDE

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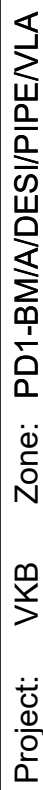


1	2	3	4	5	6	7	8
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A

Welding list see doc	-
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For installation tolerances dimensions to be checked by pipe contractor at site.
Degree of prefabrication to be decided by contractor.

A

NDT Long. weld %	-

For installation tolerances dimensions to be checked by pipe contractor at site.
Degree of prefabrication to be decided by contractor.